2 **(a)** Define acceleration.

.....

..... [1]

(b) A train has a total mass of 520 000 kg. The train accelerates at 1.1 m / s^2.

(i) Calculate the time taken for the train to increase its speed from 15 m / s to 28 m / s.

time = ... [2]

(ii) Calculate the force required to produce an acceleration of 1.1 m / s^2 for this train.

force = ... [2]

(iii) The train uses electric motors.

Explain why the force on the train due to the motors is greater than the value calculated in **(ii)**.

.....

..... [1]

[Total: 6]

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